

ENBC332: Homework #3

Due: Monday, 09/29/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 50

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g. "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Understanding the z-score concept

Problem 1. Generate a dataset of 5000 observations from a normal distribution with mean 75 and standard deviation 7. (10 Points)

Hint: Use `rnorm()` function to create the data.

a) Using `ggplot2`, create a histogram of the data with 50 bins. Add a density curve, `geom_density()`, to the plot to compare the sample distribution to the theoretical normal shape.

b) Write a function that calculates z-score. Calculate the z-scores for the following data points: 68, 75, and 96. Use **ggplot** to add vertical lines to the plot. Highlight each vertical line with different colors. Attach the plot to your homework.

Hint: You have three inputs for your function (x , μ , and σ)

c) Interpret what these z-scores mean in the context of the normal distribution.

Understanding the normal distribution concept

Problem 2. Consider a normally distributed dataset with a mean of 50 and standard deviation of 10. (20 Points)

- a) Calculate the probability that a randomly chosen observation is between 40 and 60. Use **ggplot** to highlight the area in the plot. Attach the plot to your homework.
- b) Calculate the probability that a randomly chosen observation is between 5 and 75. Use **ggplot** to highlight the area in the plot. Attach the plot to your homework.
- c) Calculate the probability that a randomly chosen observation is greater than 60. Use **ggplot** to highlight the area in the plot. Attach the plot to your homework.
- d) Calculate the probability that a randomly chosen observation is smaller than 90. Use **ggplot** to highlight the area in the plot. Attach the plot to your homework.

Understanding the binomial distribution concept

Problem 3. Suppose a clinical trial where a new drug is being tested. The probability of a patient responding positively to the drug is 0.7 (70%). The trial involves administering the drug to 20 patients.

Hint: Use `rbinom()` function to create the data with sufficient number of trails i.e. `rbinom(1000, size = 20, prob = 0.7)`.

- a) Calculate the probability that exactly 15 patients will respond positively.
- b) Calculate the probability that more than 12 patients will respond positively.
- c) Calculate the probability that between 10 and 18 patients (inclusive) will respond positively.
- d) The problem indicated that the probability of a patient responding positively to the drug is 0.7 (70%). Let's examine the distribution considering probability of a patient responding positively to the drug for 30%, 50%, and 70%. Create a histogram for each probability (30%, 50%, and 70%) to visualize the distribution of the number of successes. Attach all three histogram plots to your homework and interpret your observation from these three plots.

Hint: Similar to part a, use `rbinom()` function to create the data with sufficient number of trails i.e. `rbinom(1000, size = 20, prob = 0.7)`.